EXP 09: ANALYSIS OF PHOTODETECTOR NOISE AND SNR USING MATLAB/SCILAB

OBJECTIVES :

1. Analyze the key noise sources in an optical photodetector, namely shot noise, dark current noise and thermal (receiver) noise.

2. Compute the individual and total noise current variances (σ^2) across a range of input optical powers.

3. Evaluate the Signal-to-Noise Ratio (SNR) and observe its scaling in the thermalnoise-limited and shot-noise-limited regimes.

4. Calculate the Minimum Detectable Optical Power (Pmin) at SNR = 1 (0 dB). 5. Simulate and visualize photodetector noise and SNR curves using MATLAB.

SOFTWARE USED : MATLAB

MATLAB PROGRAM:

```
% Photodetector Noise and SNR Analysis
clc; clear;
close all;
%% 1. Constants and input parameters q = 1.602e-19;  kB = 1.381e-23;  h = 6.626e-34;
c = 2.998e8;
lambda = 1550e-9;          % wavelength (m)
eta    = 0.85;             % quantum efficiency
Id     = 5e-9;             % dark current (A)
T      = 300;              % temperature (K)
RL     = 50;               % load resistance (ohm)
Fn     = 2;                % noise figure (3 dB = 2 linear)
B      = 1e9;              % bandwidth (Hz)
Pdbm   = -60:0.1:0;        % input power sweep (dBm)
P      = 10.^((Pdbm-30)/10);   % input power (W)
%% 2. Responsivity and signal current
R  = eta*q*lambda/(h*c);
Ip = R*P;
%% 3. Noise variances and SNR
var_shot = 2*q*Ip*B; var_dark =
2*q*Id*B; var_th   =
4*kB*T*B*Fn/RL; var_tot  =
var_shot + var_dark + var_th;
SNR_dB   = 10*log10(Ip.^2 ./ var_tot);
```

%% **4. Minimum detectable power (SNR = 1)**

```
a = R^2;  b = -2*q*B*R;  cc = -(var_dark + var_th);
Pmin    = (-b + sqrt(b^2 - 4*a*cc))/(2*a);
PmindBm = 10*log10(Pmin) + 30;
```

%% **5. Display results**

```
[~,k] = min(abs(Pdbm+30));          % index of -30 dBm
fprintf('Responsivity R      = %.4f A/W\n', R);
fprintf('Thermal variance    = %.3e A^2\n', var_th);
fprintf('Dark variance       = %.3e A^2\n', var_dark);
fprintf('Shot variance(-30)  = %.3e A^2\n', var_shot(k));
fprintf('SNR at -30 dBm      = %.2f dB\n', SNR_dB(k));
fprintf('Pmin (SNR = 1)      = %.2f dBm (%.1f nW)\n', PmindBm, Pmin*1e9);
```

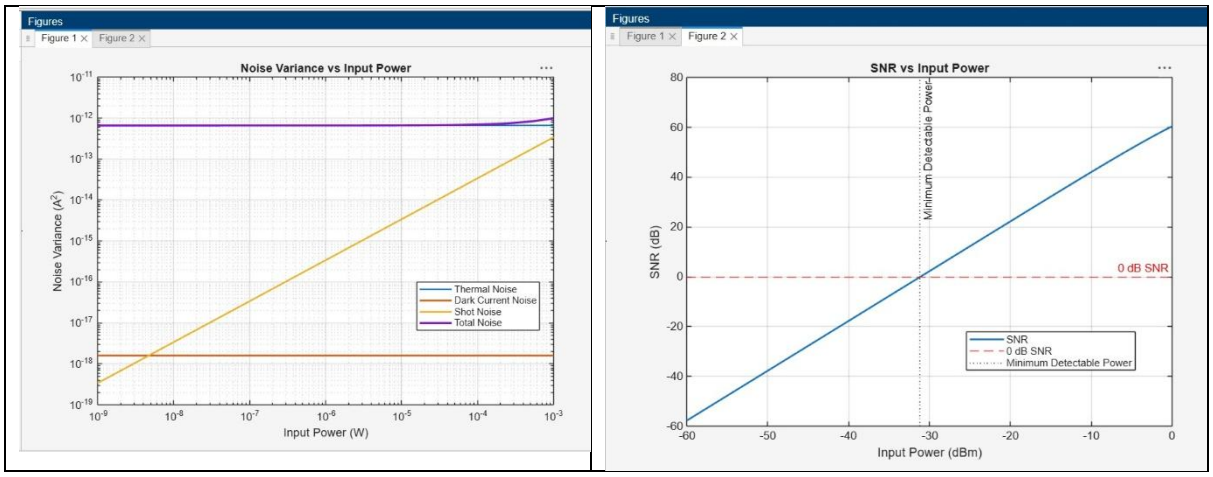
%% **6.Plots**

```
subplot(2,1,1);
loglog(P, var_th*ones(size(P)), P, var_dark*ones(size(P)), P, var_shot, P,
var_tot); grid on; xlabel('P_{in} (W)'); ylabel('Noise variance (A^2)');
legend('Thermal','Dark','Shot','Total','Location','best'); subplot(2,1,2);
plot(Pdbm, SNR_dB); hold on; plot([-60 0],[0 0],'r--'); plot([PmindBm
PmindBm],[-60
70],'k:'); grid on; xlabel('P_{in} (dBm)');
ylabel('SNR (dB)'); Expected Outputs:
```

Command Window Output

```
Responsivity R      = 1.0625 A/W
Thermal variance    = 6.629e-13 A^2
Dark variance       = 1.602e-18 A^2
Shot variance(-30)  = 3.404e-16 A^2
SNR at -30 dBm      = 2.31 dB
Pmin (SNR = 1)      = -31.16 dBm (766.4 nW)
```

RESULTS:



RUBRICS :

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Develop the MATLAB program with correct constants and system parameters.	Program Setup and Parameter Definition	Defines all constants and parameters correctly with proper units and a clean script structure, independently.	Defines parameters correctly with minor unit or syntax errors and limited assistance.	Defines most parameters but needs considerable assistance or has unit errors (e.g., dBm/W, linear F_n).	Unable to set up the script or define the parameters.	25
Compute R, signal current, noise variances and SNR across the power sweep.	Simulation Execution and Data Acquisition	Computes R, I_p, all noise variances and SNR correctly over −60 to 0 dBm and records the values.	Computes all quantities with minor errors or omissions.	Computes some quantities with noticeable errors or inconsistencies.	Unable to run the simulation or record the values.	25

Obtain the noise and SNR plots and determine P_{min}.	Graph PloBing and P_{min} Determination	Plots accurate log–log and semi-log curves with labels and reference lines; P_{min} within 5% of the reference.	Plots the curves with minor errors; P_{min} identified with small deviation.	Plots partial or poorly scaled curves; P_{min} approximate.	Unable to plot the curves or determine P_{min}.	25
Interpret noise regimes, SNR slopes and receiver sensitivity.	Analysis and Engineering Interpretation	Correctly identifies thermal- and shot-limited regimes, explains the SNR slopes and links results to receiver design.	Identifies the regimes with minor errors and reasonable interpretation.	Performs basic interpretation with noticeable errors or limited justification.	Unable to interpret the results.	25

REFLECTION :

I learned to analyze shot, dark-current, and thermal noise in a photodetector using MATLAB and calculate reposivity, SNR, and minimum detectable optical power. I observed how noiseaffects receiver sensitivity and verified that SNR increases with input optical power.

RESULT :

The study shows how input optical power affects noise and SNR in a PIN photodetector receiver. Performance depends on the power level:

• Low power: thermal noise dominates, giving negative SNR and no reliable detection (below Pmin ≈ −31.16 dBm).

• Moderate power: the signal rises above the thermal noise and the SNR increases at 20 dB/decade.

• High power: SNR becomes large and shot noise grows linearly, approaching the thermal level near +2.9 dBm.

These results help in estimating the receiver sensitivity and in designing low-noise optical receivers.